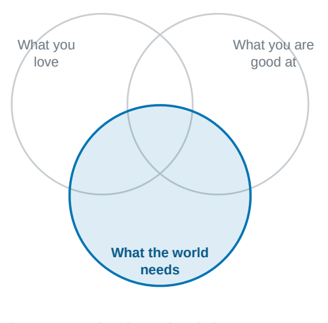
This activity is running inside Canvas. If you would prefer, you can **open the full page in its own tab ↗** and come back to Canvas to submit what you copied. Open it before you start writing, so everything you type is saved in one place.

CST349 · FALL 2026 · WEEK 2 · OWN YOUR PROGRESS · WILL NOT AFFECT YOUR FINAL GRADE

Read the Market

Due Wednesday 2 September, 11:59 p.m. · started in Wednesday's session · about 60 minutes

Take ownership of your progress — this activity is provided to help you, but the grade for this activity will not affect your final grade.



Today: what the world needs. Friday: work that holds your attention. Tuesday: the evidence to build next.

TODAY YOU WILL

- Write down a unique problem, opportunity or niche in the job market that you can search.
- Research and identify at least two things employers are asking for now, with specific terms, phrases and how often.
- Make note of at least one thing you expect to change, with its drivers and the signal you would watch.

Where this week is going

This week you connect three things: what the market is asking for, work that holds your attention, and the evidence you want to build next. Together they shape your direction this semester.

In two to three months, Prof. Sathya may send an email message about you to one of his industry contacts. You draft that message yourself in Find Your Gap, the third of this week's three activities, from the work you do in the first two. Two questions set the standard for it:

Would Prof. Sathya actually send this?
Would his contact actually be spurred to action by it?

A contact acts when they can see a specific opportunity and a specific reason you fit it — which is why the week starts with real job postings.

Purpose

Find out what employers are asking for now, through market data and job postings — and develop the skill to continuously update your understanding based on such data.

Steps

1. Understand the learning outcome
2. Identify a niche to look at
3. Read real postings
4. Evaluate against the criteria
5. Final reflection

The CST349 AI Dojo knows about this activity; asking it to 'Walk me through the Read the Market' might be helpful if you feel stuck.

If you want to see it in use first: a video walkthrough, and a shared ChatGPT conversation on this activity.

Enter Full Name First and last

Asked once, here. It goes into the summary you paste into Canvas at the end.

Your partner's full name First and last

You will work with the same partner throughout this activity.

Learning outcome

Identify a niche in the job market I can search; find at least two things employers in it are asking for now, in their own words and with a count of how often; and identify one thing I expect to change, with the driver behind it and the signal I would watch.

Two worked examples

What the outcome looks like when it is met — written by us as models, not student work.

Example 1 — six new-grad software and data postings.

The niche. Backend and data engineering roles for new graduates at consumer software companies.

Asking for now, one. Code review from day one — 3 of the 6 postings. Underdog: "participating in code reviews and ensuring engineering best practices"; Chewy: "Participate in code reviews to understand design principles, version control, and testing best practices".

Asking for now, two. Side projects listed as a qualification, next to the degree — 2 of the 6. Underdog: "Track record of shipping high-quality, feature-rich products with a portfolio of side projects". I had assumed side projects were a nice-to-have.

What I expect to change. More postings will say up front that AI screens applications before a person reads them. Today 2 of the 6 say so — Sigma Computing: "This hiring process utilizes artificial intelligence tools to assist in candidate screening and assessment". **Driver:** states are starting to require employers to tell applicants when AI is used in hiring decisions. **Signal:** in three months, how many of 10 fresh postings in this niche carry a screening disclosure.

What I would do. Use the rest of my time in college on projects with real impact, so that what I can show *is* the qualification.

Example 2 — five cybersecurity postings.

The niche. Graduate application-security and product-security roles at large consumer platforms.

Asking for now, one. Code in more than one language, as a minimum — 2 of the 5. Both TikTok graduate roles, Product Security Engineer, Seattle and Application Security Engineer, Singapore: "writing and reviewing code in at least two of the following programming languages: Kotlin, Swift, TypeScript, Go, Python, Rust" — and none of those is the language most of my coursework is in.

Asking for now, two. Public security work done on my own — 2 of the 5 list "CTF or Live Hacking Event experience" and "Bug bounty experience" as preferred qualifications, things I can do this semester without waiting for an internship.

What I expect to change. Reviewing code that an AI wrote will show up as its own line in these postings. **Driver:** more of the code these teams ship is AI-generated, and someone has to check it for the same flaws a person would introduce. **Signal:** in three months, whether any of 10 fresh postings mention reviewing AI-generated code.

What I would do. Use the rest of my time in college on projects with real impact — pick up Go or Rust properly and enter events where the work is public.

What you are building on

No prior market research is required. If you have never read a job posting closely, this is the right place to start. The reading is the activity.

Write your response first. Then use **Check it yourself**, **AI feedback**, or both. Both are under every box and they go at the same thing from two directions.

Your work is saved in this browser as you type. Nothing leaves the page until you press a feedback button or copy your summary.

Continue — identify a niche Open. You can keep working above.

Identify a niche to look at

1 · Identify a niche to look at

One sentence: a kind of work you would want to do, in a kind of company, solving a kind of problem. It can change later — you need something specific enough to search for. "Backend work on data pipelines at climate-tech companies" gives you something to type into a job board; "software engineering" returns two hundred thousand openings.

Reflection questions to consider

1. What led you to that sentence in particular? Identify one thing you already believe about this kind of work.

One sentence. A kind of work, in a kind of company, solving a kind of problem.

0 characters (about 40 before the feedback button turns on)

Get AI feedback

Check it yourself — the same criteria, without the AI

- Typed into a job board, would this return a coherent set of roles?
- Does the sentence stop being true if you swap the industry out?
- Is it a kind of work, and not a job title or a technology?

Partner pause. Swap sentences with your partner and read theirs closely. Ask each other questions until you understand the reasoning behind the answer, say what you are not convinced by, and work through the difference together. Take turns.
Then record one sentence: what you changed, what you kept, or what you still need to check.

One sentence.

Continue — read real postings Open. You can keep working above.

Read real postings

Two things worth having open while you read:

Inside Hiring with Ashby
How the top of the funnel actually works, from the people who build the applicant-tracking system.

Fireside Chat with Recruiters
Recruiters on what they do with a new-grad application, chaptered with jump links.

The rest of the reference material

Apply Within, and the market data behind the week.

Apply Within (Laszlo Bock) — the PDF.

Canaries Dashboard, Stanford Digital Economy Lab (Brynjolfsson, Chandar, Chen) — ADP payroll records, not a survey or a forecast: what actually happened to employment since November 2022, broken out by age band and AI exposure. Students in this room are the exact cell in that chart. The page says *Last updated July 22, 2026*: the two most exposed occupation groups decline since ChatGPT's introduction while the other three grow, and in the most-exposed quintile early-career women contract 4.5% per year and men 2.5% per year. It breaks out four named occupations — software developers, customer service representatives, stock clerks, home health aides — so there is no data science, no security and no game development in it.

Canaries in the Coal Mine — the August 2026 revision, Brynjolfsson, Chandar and Chen, 12 August 2026 — the paper behind the dashboard, with data through June 2026. Two things it states that the dashboard page does not. On the size of the gap: "Employment among workers ages 22–25 in highly AI-exposed occupations now stands about 19% below where it would be if it had kept pace with employment among similarly aged workers in less-exposed occupations." By that same measure the shortfall was 15% at the July 2025 data vintage and is 19% as of June 2026. And on how it is happening: "The adjustment appears to operate primarily through reduced hiring of young workers rather than increased separations".

BLS Occupational Outlook Handbook — "Software Developers, Quality Assurance Analysts, and Testers" — the ten-year national projection: 10 percent growth 2025–2035, last modified 27 August 2026.

Online Search Guidance

The searches that produced the postings quoted in the examples.

These are the searches that actually produced the postings quoted in the examples, run on 2026-09-01.

Role titles to try, not just "software engineer": `software engineer new grad 2026` · `software engineer I` · `associate software engineer` · `data engineer I` · `application security engineer graduate` · `product security engineer graduate` · `junior data scientist`.

Add where postings actually live: append `site:job-boards.greenhouse.io` or `site:jobs.lever.co` to the search, or add `myworkdayjobs.com`. This is what separates a real posting from a page about postings.

Narrow with one thing at a time: the year (`2026`), the level (`new grad`, `0-2 years`, `entry level`), a place, or a technology taken from your own sentence in section 2.

What to skip: aggregator result pages — ZipRecruiter, Glassdoor and Indeed summary listings — give you counts and salary ranges but not the posting text you need to quote. You need the employer's own page.

Market snapshot

Four things that are true about entry-level hiring right now, and what each one means for the two boxes below.

SIGNAL	WHAT IT MEANS FOR THIS ACTIVITY	SOURCE
Applications per posting roughly tripled once remote work opened national pools. Around 300 applications per hire; about 18 applicants per role get a scheduled interview.	Search inside a niche you can describe.	Ashby
Recruiters run AI at the top of the funnel for speed, and a human makes every final decision.	The words a posting repeats are the words that carry. Collect them verbatim.	Ashby
Big companies pulled back on junior hiring; startups picked it up, some running intern programmes at fifteen people.	Compare employers of different sizes, not only the ones you have heard of.	Fireside
One recruiter shortlisted 2 of 300 applicants on the strength of cover letters a person had written themselves.	Specific and researched goes further than high volume.	Fireside

2 · Two things employers are asking for right now

In your niche, with the evidence for each. Read 10 to 15 real, currently open postings from different companies, and **quote the phrases** as they are written. Count how many of the postings each phrase appears in.

Reflection questions to consider

1. Which phrase appeared in the most postings, and how many out of how many? What does its frequency tell you that a single posting could not?
2. Which of your beliefs from box 1 did the postings support, and which did they not? Quote the phrase that changed your mind.

Two things, each with the exact phrases and the count — for example: "own a service end to end", 7 of 12 postings.

0 characters (about 200 before the feedback button turns on)

Get AI feedback

Check it yourself — the same criteria, without the AI

- Are these the postings' words, or a summary of them?
- Is the count there — how many out of how many?
- Does each one appear in at least half of what you read?

3 · One thing you expect to change in the next year or two

Two parts. The driver is what is pushing the change. The signal is what you would watch to find out whether the change is really happening. Start from whatever in the postings surprised you, and ask what is pushing it. The fireside session has a worked example: big companies pulled back on junior hiring, so startups became where junior roles live.

The change, the driver behind it, and the signal you would watch.

0 characters (about 200 before the feedback button turns on)

Get AI feedback

Check it yourself — the same criteria, without the AI

- Could someone else check your signal in three months without asking what you meant?
- Does the driver identify a force — a cost, a rule, a capability, a change in who is buying?
- Are the driver and the signal two different things?

Partner pause. Put your counts side by side with your partner's and find where your two niches differ.
Then record one sentence: what you changed, what you kept, or what you still need to check.

One sentence.

Partner pause. Read your signal to your partner and ask whether they could check it in three months and get a yes or a no.
Then record one sentence: what you changed, what you kept, or what you still need to check.

One sentence.

Continue — the criteria Open. You can keep working above.

Evaluate against the criteria

REVIEW YOUR WORK

- Each of the work boxes above has a response in it.
- Box 2 quotes the postings' own words and states a count.
- Box 3 has a driver and a signal, and they are two different things.
- Which answer needs stronger evidence? Note the source or posting you will check.

Reflection

Go down the criteria. Which of your two findings is best supported by quoted evidence, and which is weakest? Why?

Your answer...

Get AI feedback

Check it yourself — the same criteria, without the AI

- Write the criterion next to each of your two findings.
- For the weaker one, name the element that is not in it yet — a quoted phrase, a count, a consequence.
- Add that element, then read it again.

Partner pause. Ask your partner which of their findings they think is weakest, and work on it with them against the criteria above.
Then record one sentence: what you changed, what you kept, or what you still need to check.

One sentence.

Continue — final reflection Open. You can keep working above.

Final reflection

Reflection

What did you learn about reading the market that you would want to use again, and when? What would you want to find out next?

Your answer...

Get AI feedback

Check it yourself — the same criteria, without the AI

- Is what you learned something you did — a way of searching, reading or counting?
- For the "when", name the occasion: a month from now, before an application, when the kind of work you are after changes.
- Your next question — could you start on it tomorrow, with the postings you already have open?

Partner pause. Walk your partner through how you searched — the terms, the sites, what you skipped — and have them walk you through theirs.
Then record one sentence: what you changed, what you kept, or what you still need to check.

One sentence.

5 · What would you change about this activity?

Anything — the questions, the order, how long it took, what was unclear, what was worth doing and what was not. This is read.

Whatever you would change.

Submit

Press the button. It gathers your responses into one block and copies them to your clipboard; paste that into the Canvas box. Any AI feedback you took comes along with it, so the course can see how the feedback is working.

Copy my summary

The text that gets copied

Optional

The sessions the snapshot came from

Both worth going through in full when you have the time.

Inside Hiring with Ashby — eight short clips, about 33 minutes, with Kevin Connolly's reference slides.

Fireside Chat with Recruiters — a chaptered 89-minute recording with two recruiters.

The parts the snapshot leaves out — how long hiring takes, what survives a thousand resume reads, negotiating, the student questions — cover things that come up later this semester.

Futures thinking, properly

The method behind the driver and signal in box 3.

Finding the force behind a change and then finding the thing you can watch is the core move in futures thinking. The Institute for the Future teaches it in Ready, Set, Future! Introduction to Futures Thinking on Coursera, part of a longer Futures Thinking specialization. Nothing on this page depends on it.

The Know the Market practice Dojo

For when you have postings open but the terms are not resolving into a pattern.

What it needs from you first: 3 to 5 real postings pasted in as text. It cannot open links or browse, so bring the actual words.

What it does. It walks you through harvesting the exact terms in five passes — titles, skills, tools, qualifications and domain, then whole phrases that repeat — one per turn. Then it groups them into a keyword set, states your segment as a one-sentence working hypothesis from a small sample, and builds a search you can click to find more postings like them.

What it does not cover: box 3. It reads the market as it is now; the driver and the signal stay yours.

Open `https://dojo.symbioticthinking.ai/?topic=know-market` — if it asks for a key, use the link from the week 1 email, subject *Symbiotic Thinking Dojo API Key*. It ends by handing you a summary block and a JSON block: those are working material to draw on for the boxes above, in your own words. What goes to Canvas is the boxes and the copy button.

Your own AI Dojo

For carrying this further than the page asks.

Your own Dojo keeps your context across the semester, which this page does not, and it is where the standing routine from Tuesday will live. A prompt to start from:

> I am an early-career computing student aiming at **[YOUR NICHE]**. Find 12 real, currently open job postings from different companies where that niche is explicit in the responsibilities or requirements. For each, give me the company, title, link, and quote the 3–5 exact phrases that describe the work. Then tell me which phrases recur across at least half of them, and show me the count. Do not invent postings — only ones you can link to.
>
> Then, separately: what has changed in postings like these over the last two years, what is driving that change, and what observable signal would tell me in three months whether it is continuing? For each claim, show me your evidence, and tell me where you are inferring.

What comes back is material for your answer, not the answer. You supplied the niche; you decide which repeated phrases matter for the work you want. Pick one of its claims and check it against a posting you open yourself.

← Set Up Your CST349 AI Dojo
Course home →

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Anything here confusing, broken, or missing? Tell me anonymously — it is the fastest way to get it fixed.

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